



MASTER TUTOR SCRIPT: MHF4U UNIT 4 - TRIGONOMETRY & RADIANS Student: Rigden Total Slides: 26

HOW TO USE THIS SCRIPT: Keep this text file open on your second monitor or print it out. The HTML presentation file is completely clean and student-safe. Guide Rigden through the interactive slides using the prompts, questions, and expected answers below.

SLIDE 1: Title Slide

[SAY] Welcome to Unit 4, Rigden! We are officially stepping into the world of Advanced Trigonometry. [SAY] You've done trig before with triangles and degrees in Grade 11. But in Grade 12, we throw degrees out the window. Today, you learn the actual mathematical language of the universe: Radians.

SLIDE 2: Today's Roadmap

[SAY] Here is our game plan. [SAY] First, we learn Radian Measure. Second, we unlock the Unit Circle and learn how to find exact values. Third, graphing all 6 trigonometric waves. Fourth, transformations. And finally, real-world applications.

SLIDE 3: Degrees vs. Reality

[SAY] Look at this slide. Since childhood, you were told a circle has 360 degrees. [ASK] But why 360? Why not 100 or 1000? [EXPECT] Maybe because of a calendar or something? [SAY] Exactly! The ancient Babylonians modeled it after their ~360 day calendar. It has absolutely no mathematical connection to the actual geometry of a circle. We need a better measurement.

SLIDE 4: Enter the Radian

[SAY] A Radian is a real, physical measurement. Look at the animation on the screen. If you peel the radius 'r' off and wrap it along the curve, the angle that makes is exactly 1 Radian. [ASK] Think about the circumference formula ($C = 2 * \pi * r$). How many times do you think a radius can wrap around a full circle? [EXPECT] $2 * \pi$ times? [SAY] Brilliant! A full circle is exactly 2π radians.

SLIDE 5: The Golden Conversion

[SAY] If a full circle is 2π radians, then a half-circle is exactly 1π radians. [SAY] This is our golden conversion ratio: 180 degrees = π radians. [ASK] If 180 degrees is π , what is 90 degrees in radians? [EXPECT] Half of π , so $\pi/2$. [SAY] Spot on. To convert mathematically, just multiply by $(\pi / 180)$.

SLIDE 6: Practice - Degrees to Radians

[SAY] Let's try converting from degrees to radians, keeping the exact fraction. [ASK] Take 120 degrees. Multiply by $(\pi / 180)$. What does that simplify to? [EXPECT] Divide by 60... gives $2\pi/3$. [ASK] Excellent. Now try 330 degrees. [EXPECT] Divide by 30... gives $11\pi/6$.

SLIDE 7: Practice - Radians to Degrees

[SAY] Now let's go backward. To convert to degrees, we multiply by $(180 / \pi)$. Notice how the π symbols cancel out! [ASK] Try $3\pi/4$. [EXPECT] $3 * 180 / 4 = 135$ degrees. [ASK] Perfect. Try $5\pi/6$. [EXPECT] $5 * 180 / 6 = 150$ degrees.

SLIDE 8: Arc Length Formula

[SAY] Because radians are a physical measurement of the radius, finding the length of a curve (arc length) becomes incredibly easy. [SAY] The formula is just $a = r * \theta$. But there is one massive trap. [TIP] Point to the red box on the slide. "The angle θ MUST be in radians. If you plug degrees into this formula, your answer will be totally wrong!" [ASK] Let's test this. If radius is 10cm and angle is $\pi/2$, what is the arc length 'a'? [EXPECT] $a = 10 * (\pi/2) = 5\pi$ cm.

SLIDE 9: Co-terminal Angles

[SAY] Angles are co-terminal if they share the exact same terminal arm. Meaning, they look identical on a graph, but one has spun around extra times. [SAY] You find them by adding or subtracting a full circle, which in radians is 2π . [ASK] If I have an angle of $\pi/4$, and I add a full 2π spin, I need a common denominator. 2π becomes $8\pi/4$. What's the new angle? [EXPECT] $9\pi/4$.

SLIDE 10: Special Triangles ($\pi/4$)

[SAY] To find exact values without a calculator, we need our Grade 11 special triangles, but updated to radians. [ASK] Look at the first triangle. Instead of 45 degrees, what radian measure do we use? (Hint: $180 / 4$). [EXPECT] $\pi/4$. [SAY] Exactly. The ratios (1, 1, root 2) never change.

SLIDE 11: Special Triangles ($\pi/6$ & $\pi/3$)

[SAY] The 30-60-90 triangle gets updated too. 30 degrees is $\pi/6$, and 60 degrees is $\pi/3$. [TIP] Remember the hint: The smallest angle ($\pi/6$) is always opposite the smallest side (1). [SAY] We use these triangles to find exact values, like $\sin(\pi/6) = 1/2$.

SLIDE 12: The Unit Circle & CAST Rule

[SAY] Here is the secret to the Unit Circle. Cosine is your x-value, and Sine is your y-value. (cos, sin). [SAY] Because of this, their signs change depending on what quadrant you are in. We use CAST to remember which ratio is positive. Start in bottom right (Q4) and go counter-clockwise: C-A-S-T. [ASK] In Quadrant 2 (S), what does that tell us? [EXPECT] Only Sine is positive. Cosine and Tangent are negative.

SLIDE 13: Evaluating Exact Values (Steps)

[SAY] Let's find the exact value of $\sin(5\pi/6)$ without a calculator. We always use a 3-step process. [SAY] Step 1: Quadrant. $5\pi/6$ is in Quadrant 2. [SAY] Step 2: CAST rule. In Q2, Sine is positive. So our answer will be positive. [ASK] Step 3: We are $1\pi/6$ away from the x-axis. Look at the $\pi/6$ triangle from earlier. What is Sine (opp/hyp) of $\pi/6$? [EXPECT] $1/2$. [SAY] Yes! So $\sin(5\pi/6) = +1/2$.

SLIDE 14: Practice: Evaluating Exact Values

[SAY] Your turn. Evaluate $\cos(4\pi/3)$. Walk me through the 3 steps. [ASK] Step 1: What quadrant is $4\pi/3$ in? [EXPECT] Just past $3\pi/3$, so Quadrant 3. [ASK] Step 2: CAST rule in Q3. Is cosine positive or negative? [EXPECT] Negative (T is positive). [ASK] Step 3: The reference angle is $\pi/3$. What is $\cos(\pi/3)$? [EXPECT] $1/2$. So the answer is $-1/2$.

SLIDE 15: Working Backwards (Finding Theta)

[SAY] Now let's go in reverse. Find all angles between 0 and 2π where $\sin(\theta) = -1/2$. [SAY] Step 1: Ignore the sign. When is $\sin = 1/2$? We know from our triangle that happens at a reference angle of $\pi/6$. [ASK] Step 2: Sine is NEGATIVE. Which two quadrants does that happen in according to CAST? [EXPECT] Quadrants 3 and 4. [SAY] Exactly. Now draw your $\pi/6$ reference angles in Q3 and Q4. In Q3, it's $\pi + \pi/6 = 7\pi/6$. In Q4, it's $2\pi - \pi/6 = 11\pi/6$.

SLIDE 16: Reciprocal Trig Ratios

[SAY] Beyond SOH CAH TOA, what if we flip the fractions upside down? We get Cosecant (csc), Secant (sec), and Cotangent (cot). [TIP] Point out the red box. "Don't get confused by the letters. 'S' goes with 'C'. Sine flips to Cosecant. Cosine flips to Secant." [ASK] If I asked you to evaluate $\sec(\pi/4)$, what is your first step? [EXPECT] Find $\cos(\pi/4)$ first, then flip the fraction!

SLIDE 17: The Base Waves (Sine & Cosine)

[SAY] We've looked at angles. Now let's graph them as functions. Imagine taking the y-values from the Unit Circle and plotting them as you spin around. [SAY] Look at the graphs. Sine starts at 0. Cosine starts at its maximum peak of 1. Both repeat every 2π .

SLIDE 18: Graphing Tangent

[SAY] Tangent is completely different. Since $\tan(x) = \sin(x) / \cos(x)$, what happens mathematically when $\cos(x)$ is 0? [EXPECT] You divide by zero. It's undefined. [SAY] Exactly! That creates Vertical Asymptotes. Look at the graph. Tangent creates these repeating curves trapped between asymptotes at $\pi/2$, $3\pi/2$, and so on. Its period is only π .

SLIDE 19: Graphing Reciprocals (e.g., Cosecant)

[SAY] Now for the Grade 12 addition: Graphing Cosecant, Secant, and Cotangent. [SAY] Look at the original wave (like Sine in blue). Wherever Sine crosses the x-axis ($y=0$), its reciprocal (Cosecant) will equal $1/0$. [ASK] What happens on a graph when you have 1 divided by 0? [EXPECT] Vertical Asymptote! [SAY] Exactly! The graphs of Csc and Sec look like repeating orange U-shapes trapped between those invisible red walls, bouncing off the sine wave.

SLIDE 20: Transformations & The 'k' Value

[SAY] Just like polynomials, we can stretch, shift, and move these waves using a, k, d, and c. [TIP] Point to the Period formula. "The 'k' value dictates the Period. But trap: The period is NOT 'k'. You must divide 2π by k." [ASK] If $k = 2$, what is our new period? [EXPECT] $2\pi / 2 = \pi$. [SAY] Right. The wave is horizontally squeezed and completes a full cycle twice as fast!

SLIDE 21: The Mapping Rule

[SAY] To actually graph a complicated equation, trying to shift the wave visually is a nightmare. Instead, we use the Mapping Rule. [SAY] You take the 5 key points of a base wave (like 0, $\pi/2$, π , $3\pi/2$, 2π) and run their coordinates through this formula: $(x/k + d, ay + c)$. [SAY] This calculates exactly where the new points land.

SLIDE 22: Your Turn! (Transformations)

[SAY] Let's test this out. Grab your paper. [ASK] Tell me the amplitude, axis, phase shift, and period for this equation: $y = -3 \cos(0.5(x - \pi)) + 1$. Watch out for the negative sign on amplitude! [EXPECT] Amp is 3 (never negative). Axis is $y=1$. Shift is right π . Period is $2\pi / 0.5 = 4\pi$. [SAY] Excellent. That negative 3 just means the cosine wave starts at a minimum instead of a maximum.

SLIDE 23: Modeling the Real World

[SAY] Why do we care about waves? Because they model Ferris wheels, tides, pendulums, and daylight hours. [SAY] To build an equation from a word problem, you just need the Max and Min values. [SAY] The Axis (c) is the average. The Amplitude (a) is the distance from axis to peak. The period gives you 'k'.

SLIDE 24: Your Turn! (Ferris Wheel)

[SAY] Let's put this into practice. Calculate the Axis, Amplitude, and k-value for this Ferris wheel. Max is 20, Min is 2. Period is 40s. [EXPECT] Axis (c) = 11. Amp (a) = 9. $k = 2\pi/40 = \pi/20$. [ASK] Great. Now, because you start boarding at the bottom (a minimum), what happens to the Cosine equation? Do we need a phase shift? [EXPECT] We use a NEGATIVE cosine, so no phase shift is needed! [SAY] Spot on. The equation is $y = -9 \cos(\pi/20 t) + 11$.

SLIDE 25: Applications: Ocean Tides

[SAY] Tides work the same way. Let's look at this one. High tide is 12m at midnight ($t=0$). Low tide is 2m at 6 AM. [SAY] Notice they gave us half the cycle! High to Low is half a wave. So the full period is 12 hours. [ASK] What is the k-value? [EXPECT] $2\pi / 12 = \pi/6$. [SAY] Right. And since it starts at a maximum at $t=0$, we use a positive cosine curve.

SLIDE 26: Unit 4 Consolidation (Exit Ticket)

[SAY] You dominated this session, Rigden. We covered the entire Unit 4 curriculum. Let's do a rapid-fire exit ticket to lock it in. [ASK] 1. Convert 90 degrees to exact radians. (Expected: $\pi/2$) [ASK] 2. What is the period of $y = \sin(4x)$? (Expected: $2\pi/4 = \pi/2$) [ASK] 3. Where does Tangent (tan) have vertical asymptotes? (Expected: $\pi/2, 3\pi/2...$). [SAY] Amazing. Next time, we'll dive into the Practice Mastery deck to test these skills!